

CISC 5001 Senior Project Proposal

Candidate: Zachery Francis

Proposed Faculty Supervisor: Professor Zhou

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Good afternoon, Professor Zhao,

Hope everything is well with you. My name is Zachery Francis, and I enrolled in your class CISC 3115 in Fall 2025. I am writing this letter to ask if you will be able to act as my designated CIS Dept supervisor for CISC 5001.

Project Background

The proposed project is built on Popular Vote, which is an open-source web application that I have built along with my fellow students as part of the course 'Intro to Software Engineering' offered by Shopify's Senior SWE, Alec Malstrom. The source code is available at the following URL:

<https://github.com/Marqed4/PopularVote>, while the live deployment is available at <https://popularvote.marqed.it>. (please allow static/web services time to wake up) The application offers privacy in the query sessions and employs AI-based clustering techniques for analysis of user responses.

Proposed Research & Extensions

For my senior project, I aim to build upon the work done on the Popular Vote by adding a client-side hashing pipeline that will allow the user to generate his or her own hashing algorithm using primitives like XORing, bit rotating, FNV hashing, and salting. This will be done completely in-browser before any voting or identity information is sent to the server. I will also move the back-end code to Python to add more analysis features using NumPy and pandas libraries.

Primary Research Questions

This leads directly to the central research problem dealing with the essential trade-off between cryptographic flexibility defined by the user and cryptographic hardness. While remaining faithful to the rule of "never roll your own crypto," I will explore the trade-offs resulting from modular combinations of already proven cryptographic primitives. In particular, I am interested in the following: can the construction of a user-configured pipeline be reconciled with Kerckhoff's principle, or will the dynamic composition of primitives such as XORs and bit-rotations merely amount to "security through obscurity"?

While standardized cryptographic constructions like SHA-256 or Argon2 have been thoroughly analyzed by researchers and cryptographers, user-configured pipelines certainly have not enjoyed this luxury. My goal is to position custom pipelines along the security spectrum and to analyze the impact for privacy software. Moreover, as the back-end clustering algorithm relies on structured input, the second goal of the project is to determine the mathematical limit for hashing client-side data in order to preserve usefulness for statistical analysis.

Project Timeline

The planned schedule involves building the hash pipeline and moving to the back end in the fall semester, after which testing, analysis, and delivering the finished cross-platform project takes place in the next semester.

Key Areas of Exploration

Key areas I hope to explore under your supervision include:

- **Client-side** hashing pipeline design within the existing Popular Vote architecture
- **Backend** migration and statistical analysis of anonymized voting data
- **Flexibility** vs. cryptographic hardness framed through Kerckhoffs' principle
- **Vote** integrity and client-side privacy guarantees in real-world voting contexts

I would greatly value your guidance and constructive criticism, especially with regard to any weaknesses in this method that might be more vulnerable than originally thought. I am glad to write out an extensive proposal for you and meet to discuss the entire thing at your convenience.

Thank you for your time and consideration.

Have a wonderful summer,

Zachery Francis